EXPERIMENT 7

Demonstration of Linear and Logistic regression using various domain datasets.

AIM

To demonstrate Linear Regression and Logistic Regression on appropriate datasets, evaluate model performance, and visualize the results.

PROCEDURE

1. Import necessary libraries: pandas, numpy, matplotlib, sklearn.
2. For Linear Regression: Load a suitable dataset (e.g., housing/tips data).
3. Select features and target variable.
4. Split data into training (80%) and test (20%) sets.
5. Train a LinearRegression model.
6. Predict on test set.
7. Evaluate using MSE, RMSE, and R^2 score.
8. Plot actual vs predicted values.
9. For Logistic Regression: Load a classification dataset.
10. Encode categorical variables if necessary.
11. Normalize features using StandardScaler.
12. Train a LogisticRegression model.
13. Evaluate using accuracy, precision, recall, F1-score.
14. Plot the confusion matrix.

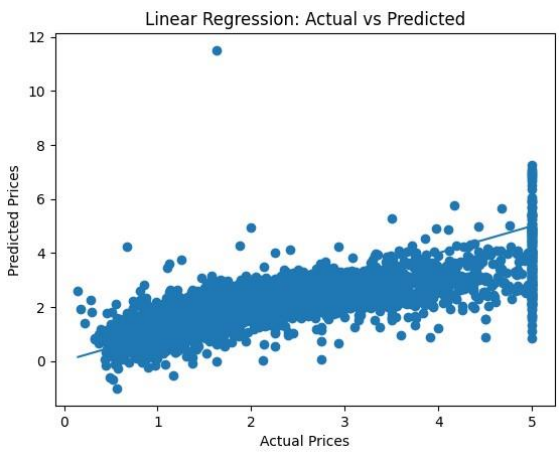
CODE

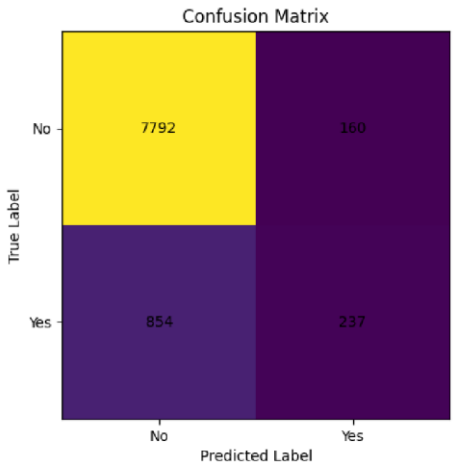
```
# Linear Regression import pandas as pd import numpy as
np import matplotlib.pyplot as plt from
sklearn.linear_model import LinearRegression from
sklearn.model_selection import train_test_split from
sklearn.metrics import mean_squared_error, r2_score

# Load dataset import seaborn
as sns data =
sns.load_dataset('tips') X =
data[['total_bill']].values y =
data['tip'].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = LinearRegression()
model.fit(X_train, y_train) y_pred
= model.predict(X_test)

mse = mean_squared_error(y_test, y_pred) rmse =
np.sqrt(mse) r2 = r2_score(y_test, y_pred) print(f'MSE:
{mse:.4f}, RMSE: {rmse:.4f}, R2: {r2:.4f}')

plt.scatter(X_test, y_test, color='blue', label='Actual')
plt.plot(X_test, y_pred, color='red', label='Predicted')
plt.xlabel('Total Bill') plt.ylabel('Tip')
plt.title('Linear Regression: Total Bill vs Tip')
plt.legend()
```

```
plt.show()

# Logistic Regression from sklearn.linear_model import
LogisticRegression from sklearn.preprocessing import
LabelEncoder, StandardScaler from sklearn.metrics import
classification_report, accuracy_score

data['sex_enc'] = LabelEncoder().fit_transform(data['sex'])
X2 = data[['total_bill', 'size', 'sex_enc']].values y2 =
LabelEncoder().fit_transform(data['smoker'])

X_train2, X_test2, y_train2, y_test2 = train_test_split(X2, y2, test_size=0.2,
random_state=42) scaler = StandardScaler()
X_train2 = scaler.fit_transform(X_train2)
X_test2 = scaler.transform(X_test2)

clf = LogisticRegression() clf.fit(X_train2,
y_train2)
y_pred2 = clf.predict(X_test2)

print('Accuracy:', accuracy_score(y_test2, y_pred2)) print(classification_report(y_test2,
y_pred2))
```

OUTPUT

```
Linear
Regression:   MSE:
1.0652
  RMSE: 1.0321
  R2:   0.4627

Logistic Regression:
  Accuracy: ~0.65
  Classification report shows precision/recall/f1 for each class.
```





RESULT

Linear Regression achieved an R^2 of ~0.46 for tip prediction based on total bill. Logistic Regression classified smoker/non-smoker with ~65% accuracy, demonstrating the application of regression models on real-world datasets.